17 x is directly proportional to the square of y.
y is directly proportional to the cube of z.

$z = 2$ when $x = 32$

Find a formula for x in terms of z.

$x = ky^2$ and $y = cz^3$

Therefore $x = k(cz^3)^2 = Kz^6$

At $z=2$, $x=32$: $32 = K \times 2^6 = 64K$

$K = 1/2$, so $x = z^6/2$

1 mark
x proportional to y^2

1 mark
eliminate y: z^6

1 mark
constant from 32,2

1 mark
$x=z^6/2$

$z^6 / 2$

(Total for Question 17 is 4 marks)

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